

cTEE: Cryptographic Trusted Execution Envelopes

Private Workspaces and Protected Agency on Untrusted Compute

cTEE Yellow Paper

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Abstract

Modern agent harnesses increasingly need persistent remote compute: rented GPUs, hosted workers, decentralized compute nodes, service marketplaces, and always-on cloud workspaces. The standard deployment model gives the remote node a plaintext workspace and then asks containers, VMs, process sandboxes, or boot measurements to keep a root-level provider from reading it. That model is wrong for high-value agent state. A provider with root can usually inspect files, logs, process memory, prompts, tool inputs, and GPU-visible buffers.

This paper introduces cTEEs, Cryptographic Trusted Execution Envelopes, as a software systems pattern for private autonomous agents on untrusted compute. A cTEE is not a hardware TEE and does not claim to hide arbitrary plaintext memory from root. Its central invariant is stricter: protected workspace state is not given to the untrusted node as normal plaintext in the first place. The user-facing primitive is an *Open Private Workspace*; the node-facing primitive is a cryptographic execution envelope over encrypted workspace objects, public trunks, sealed private heads, commitments, receipts, leakage profiles, declassification gates, and capability exits. A cTEE can be bound to any private agent harness, enterprise runner, DePIN network, worker runtime, or service marketplace that can provide an authority view, execution boundary, state log, artifact store, and optional settlement layer.

The paper contributes: (1) a formal no-plaintext-custody model for private agent workspaces; (2) a concrete object and protocol suite including `PrivateWorkspaceNode`, `PrivateWorkspaceCapsule`, `AlphaSeal`, `AutonomyLease`, `DeclassificationReceipt`, and `PrivateInferenceReceipt`; (3) a compositional security treatment for privacy, authority, restore validity, and receipt binding; and (4) a practical default high-throughput execution strategy, Candidate-Lattice Private Decoding, that keeps ordinary GPU token speed for public model work while moving private selection, private policy, or private strategy heads outside the untrusted node. It also defines Plaintext-Free Runtime Mounting, including Plaintext-Free Model Mounting, as the runtime interface that exposes private workspaces to runtimes as public/redacted content, encrypted refs, private handles, and capability exits rather than provider-readable plaintext. We also state a negative result: arbitrary full-speed private LLM inference on a single root-controlled consumer GPU is not obtained by software wrapping alone. Useful privacy requires either no plaintext custody, trusted hardware, a trusted or threshold guardian, or cryptographic evaluation with its own overhead. cTEEs make that tradeoff explicit and productizable.

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1 Motivation

Private autonomous agents need compute that can stay awake. A developer may want to rent a cheap RTX 3090 node to run a persistent agent harness. A quant user may want the node to fetch market data, run backtests, simulate models, draft reports, and propose trades while the user’s laptop is asleep. An enterprise may want a service marketplace to spin up decentralized compute for a private outcome. In each case the economic reason to use the node is clear: GPU time is cheaper or more available away from the user’s local machine.

The privacy problem is equally clear. If the remote workspace contains:

```
/workspace/strategy.py  
/workspace/signals.py  
/workspace/risk.py  
/workspace/config.yaml
```

as ordinary plaintext, a root-level provider can likely copy the files, snapshot the disk, trace the process, inspect logs, and watch prompts. An encrypted disk image helps at rest, but once the guest mounts the volume and the agent reads the strategy, the provider-controlled host can often observe the plaintext through some layer of the system. A container boundary is not a privacy theorem.

cTEE doctrine. The node can compute against the workspace, but it should not custody workspace secrets.

1.1 Why This Is Not Encryption at Rest

Encryption at rest protects stored bytes until a program decrypts them. It does not, by itself, protect the plaintext once the remote workspace mounts the volume, prompts a model, runs a tool, writes a log, caches a result, or places intermediate values in provider-visible memory.

cTEE changes the routing contract. Protected state is not mounted, prompted, logged, cached, traced, or executed as provider-readable plaintext on the untrusted node. The node receives public trunks, redacted projections, encrypted refs, commitments, candidate lattices, and receipts. Plaintext views and declassification decisions live in the user’s authority view.

cTEEs are proposed as the missing middle between two bad extremes:

1. **Plain cloud workspace.** Fast and useful, but the provider can read the secrets.
2. **Pure cryptographic private inference.** Stronger privacy, but today usually too slow or too narrow for arbitrary agentic LLM work.

cTEEs keep the useful fast path by splitting work into public or redacted trunks that can run on the rented GPU, and private heads that remain sealed, masked, secret-shared, guardian-mediated, or locally evaluated.

2 Contributions

This yellow paper defines cTEEs as a systems primitive, not a new cryptographic primitive. Its technical contribution is a custody discipline for private agency: a way to transform many secret-bearing agent tasks into public candidate expansion, private choice, receipted declassification, and capability-bounded action.

1. **No-Plaintext-Custody Security.** We formalize the property that an untrusted node’s view is simulatable from declared leakage, public trunks, encrypted refs, commitments, and receipts.
2. **Private Workspace Object Model.** We define the objects needed to make “Open Private Workspace” real: encrypted workspace objects, private workspace capsules, sealed private heads, autonomy leases, declassification receipts, and capability exits.
3. **Split-State Agent Automaton.** We model agent execution as a state machine whose state is partitioned into public, redacted, sealed, guardian, and action-authority components.
4. **Custody Type System.** We define a small type discipline for plaintext custody. Values are typed as public, redacted, sealed, guardian-only, cryptographic-operator, capability-only, or never-remote-plaintext, and runtime mounts must type-check before node exposure.
5. **Proof-Carrying Workspace.** We define a verifiable custody proof that binds sensitivity labels, mount graph, operator routing, receipts, leakage budget, and state roots. A result can carry evidence that protected values did not enter node plaintext custody.
6. **Private Agency Transform.** We formalize the reduction from private inference to public proposal plus private selection for candidate-selection-reducible tasks. This is the theorem-shaped reason cTEEs can preserve node-side token speed for useful private agency without claiming arbitrary private LLM inference.
7. **Candidate Coverage Frontier.** We show that zero online private-selection leakage can be purchased with public overgeneration in proportion to candidate redundancy mass. When acceptable complete traces retain non-negligible public probability mass, the trace budget is depth-independent; when that mass decays exponentially, cTEE must route away from CLPD/CLE.
8. **Counterfactual Lattice Execution.** We introduce a high-assurance CLPD schedule in which the node expands a committed lattice of plausible futures before learning any private selection. The private path is hidden inside counterfactual public work, replacing step-by-step selection leakage with final declassification leakage at the cost of extra public token volume.
9. **External Model API Boundary.** We distinguish cTEE no-plaintext-custody from provider trust. A third-party model API that receives sensitive plaintext is outside the base cTEE custody claim unless the payload is public, redacted, declassified, or protected by a separately verifiable private-compute interface.
10. **AlphaSeal.** We define a reusable sealed-private-head pattern for quant strategies and other high-value decision logic.
11. **Candidate-Lattice Private Decoding.** We introduce the default cTEE protected-agency strategy: an untrusted GPU generates public candidate lattices at normal speed while a private head selects, filters, or reranks without revealing the private policy.
12. **Plaintext-Free Runtime Mounting.** We define the runtime mount boundary that presents private workspaces to tools, files, shells, model servers, and model calls as public trunks, redacted projections, encrypted refs, private-function handles, declassification requests, and capability exits rather than plaintext private files or prompts. `PlaintextFreeModelMount` is the model-facing specialization.

13. **Cryptographic Operator Plane.** We define the internal operator plane that routes protected subcomputations to FHE, MPC, garbled circuits, ORAM, local guardian, or threshold guardian paths. The default second logical party is the user’s existing authority surface, not a new managed committee.
14. **Deterrence and Detection.** We specify synthetic canaries, provider-bound watermarks, honeytokens, leak scans, replay checks, and receipts as attribution and dispute evidence. These mechanisms do not make plaintext safe; they make abuse more detectable.
15. **Receipt and Restore Composition.** We specify how receipts, commitments, encrypted blobs, and state roots make private agent work auditable and restorable without exposing plaintext to the compute provider.

2.1 Security Claims and Non-Claims

Claims	The node does not receive plaintext custody of protected workspace objects. Private objects are encrypted, committed, redacted, guardian-mediated, or handled through explicit capability exits. External effects require authority and receipts. Restore/import is state-root and policy checked.
Non-claims	cTEE does not claim arbitrary full-speed private LLM inference by software wrapping alone. It does not eliminate all metadata, timing, size, schedule, or selection leakage. It does not protect a workspace that the user explicitly mounts on the rented node as ordinary plaintext. It does not turn a third-party model API that receives sensitive plaintext into a no-plaintext-custody system.

2.2 External Model API Boundary

The cTEE privacy claim is a custody claim. It is strongest when the user, service, or organization controls the compute substrate: local GPU, rented GPU, customer VPC node, DePIN node, or another venue where protected state can stay outside provider-readable plaintext custody.

If sensitive plaintext is sent to a third-party model API, the run leaves the base cTEE no-plaintext-custody model for that data and enters a provider-trust model. Contractual no-training terms, retention controls, enterprise privacy policies, and legal commitments may be useful, but they are not the same claim as never giving the provider sensitive plaintext.

- **Private-native.** Sensitive plaintext does not leave cTEE custody. Models run locally, on rented/self-hosted compute, in customer-controlled venues, or through private operators.
- **Redacted-API.** A third-party API receives only public data, redacted projections, synthetic/counterfactual candidates, or explicitly declassified payloads.
- **Provider-trust.** A third-party API may receive sensitive plaintext under contract, policy, retention promises, attestation, or other trust claims. This is outside base cTEE no-plaintext-custody unless the API exposes a separately verifiable private-compute interface.
- **Unsafe.** Private data enters untrusted plaintext custody without explicit declassification or accepted provider-trust policy.

This distinction is economic as well as cryptographic. If a user rents a GPU node, extra candidate or counterfactual generation consumes leased GPU time and wall-clock latency. If the same strategy is implemented through third-party API tokens, extra private-preserving branches may burn API budget and may also expose plaintext unless the API path is public, redacted, declassified, or separately protected.

3 System Model

3.1 Participants

A cTEE deployment has the following roles:

USER. The human, organization, or local client that owns private workspace state.

NODE. The untrusted rented, hosted, provider, or DePIN compute node. It may have a GPU and may run an agent harness. It is assumed provider-rooted.

GUARDIAN. A trusted, threshold, or policy-controlled authority path that can hold keys, evaluate narrow private heads, approve declassification, or sign capability exits. This may be a browser/device, mobile signer, enterprise key service, MPC committee, HSM, or confidential environment.

STATELOG. The append-only or operation-backed state log that records workspace refs, commits, receipts, state roots, policy hashes, and restore metadata. The state log is the canonical meaning layer for sealed objects, not the place where all payload bytes must live.

BLOBSTORE. The blob storage plane. It holds encrypted payload bytes, sealed archives, traces, checkpoints, public datasets, and large artifacts. It does not define what those bytes mean.

3.2 Guardian as Authority View

The guardian is a formal role, not necessarily a separate machine the user must operate. In the common product path, the guardian is the user's existing authority view:

- an authenticated browser instance that renders decrypted private files;
- a desktop agent harness or CLI holding an interactive workspace session;
- an authority plane or signing path backed by passkeys, hardware keys, mobile approval, policy services, or threshold custody;
- a customer VPC service, enterprise HSM, or policy service for teams;
- a small always-on authority worker for unattended **AutonomyLease** checks.

Thus cTEE does not require a second runtime beside the rented GPU node. It separates compute custody from view/sign/declassify custody. The user already needs some surface that can decrypt and display private workspace content; the guardian names that surface when it also evaluates private heads, applies declassification policy, or approves capability exits.

Authority-view doctrine. The rented node is the compute workspace. The browser, authority plane, local harness, CLI, or threshold authority is the private view and signing membrane.

3.3 Architecture at a Glance

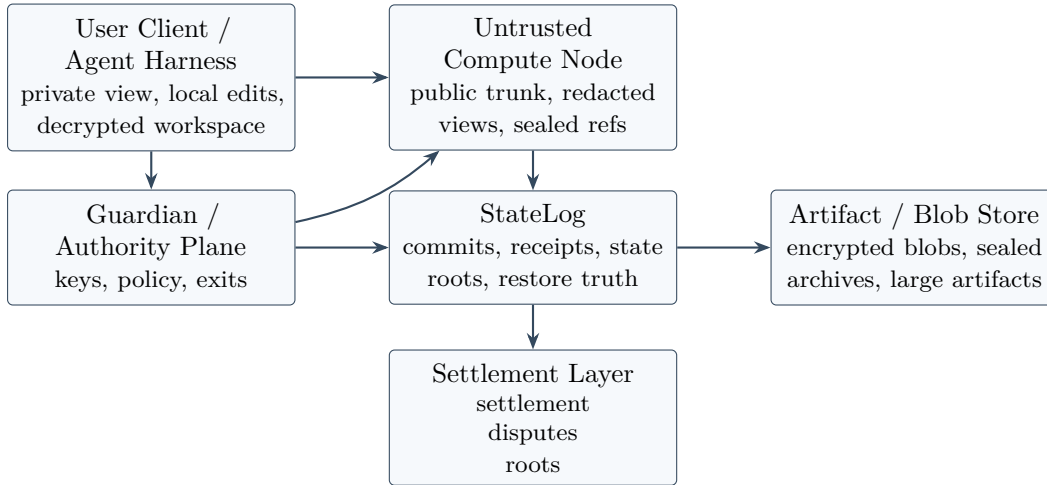


Figure 1: cTEE splits plaintext custody from compute custody. The rented node receives useful work but not protected workspace plaintext. The state log records meaning, receipts, policy, and restore truth; artifact stores hold bytes; public settlement is sparse and trigger-based.

3.4 Reference Bindings

cTEE is ecosystem-neutral. A deployment binds the generic roles above to its own runtime, authority, state, storage, marketplace, and settlement components. The protocol does not require Agentgres, wallet.network, IOI L1, or any particular marketplace. Those names are one concrete IOI binding:

Guardian / authority view.

Authenticated browser, local Autopilot, CLI signer, wallet.network-backed authority.

Authority plane.

wallet.network for grants, scopes, revocation, secrets, payments, and decryption leases.

Execution boundary.

IOI daemon for tool/model execution, sensitive-exit trapping, and runtime mount enforcement.

Plaintext-Free Runtime Mount.

Default Harness Profile cTEE mount inside the daemon.

StateLog.

Agentgres for admitted refs, receipts, state roots, policy hashes, and restore truth.

Artifact / blob store.

Agentgres Artifact Plane plus storage backends such as local disk, S3, Filecoin, CAS/IPFS, and object stores.

Settlement layer.

IOI L1 or compatible app chains for public, economic, dispute, or cross-domain commitments only when triggered.

Worker marketplace.

aiagent.xyz for discovery, manifests, benchmarks, reputation, and licensing. It advertises capabilities; it does not grant capabilities.

Outcome marketplace.

sas.xyz for service ordering, delivery, acceptance, dispute, remediation, and settlement profile. Authority still exists through the daemon and wallet.network.

Binding doctrine. cTEE defines the privacy and custody envelope. A concrete stack binds that envelope to an authority plane, execution boundary, state log, artifact store, and settlement layer. Marketplaces may advertise or contract for capabilities and outcomes, but they do not grant capabilities.

3.5 Adversary

The adversary controls NODE. It can:

- read node files, logs, environment variables, process memory, and normal GPU-visible buffers;
- snapshot disks and RAM;
- observe network timing, job sizes, model family, public datasets, and resource usage;
- delay, abort, replay, fork, or equivocate about node execution;
- return invalid outputs or try to get paid for fake work.

The adversary does not break standard cryptographic assumptions, does not compromise GUARDIAN unless the chosen guardian threshold permits it, and does not force the user to explicitly declassify protected state. Hardware TEEs may be used as an optional strengthening layer, but the base cTEE model does not depend on them.

3.6 State Partition

Let an agent workspace state S be partitioned as:

$$S = S_{\text{pub}} \parallel S_{\text{red}} \parallel S_{\text{seal}} \parallel S_{\text{guard}} \parallel S_{\text{auth}}.$$

- S_{pub} : public files, public model weights, public data, public tools.
- S_{red} : redacted summaries, public stubs, sanitized prompts, padded labels.
- S_{seal} : encrypted private workspace objects, sealed strategy heads, encrypted archives, commitments.
- S_{guard} : state available only to GUARDIAN or a threshold committee, such as key shares, private coefficients, private selectors, or declassification policy.
- S_{auth} : action authority, including signing keys, broker capability scopes, payment authority, and revocation state.

The core cTEE invariant is:

$$\text{View}_{\text{Node}} \subseteq f(S_{\text{pub}}, S_{\text{red}}, \text{Enc}(S_{\text{seal}}), \text{Com}(S_{\text{guard}}), \mathcal{L})$$

for declared leakage $\mathcal{L}$. The node’s view must not require plaintext S_{seal} , S_{guard} , or S_{auth} .

4 Cryptographic Preliminaries

4.1 Authenticated Encryption

Workspace objects are encrypted with an authenticated encryption scheme:

$$C \leftarrow \text{AEAD.Enc}_K(N, A, M), \quad M \leftarrow \text{AEAD.Dec}_K(N, A, C).$$

The associated data A should bind workspace identity, object id, schema version, policy hash, content role, state root, and expected parent head. This prevents ciphertext transplant across workspaces or policy epochs.

4.2 Commitments

A commitment scheme provides hiding and binding commitments. For scalar or vector values, a Pedersen-style commitment has the form:

$$C = rH + \sum_i v_i G_i,$$

where r is a random blinding factor and G_i, H are independent generators. It is additively homomorphic:

$$C(\mathbf{v}, r) + C(\mathbf{w}, s) = C(\mathbf{v} + \mathbf{w}, r + s).$$

This is useful in cTEEs for private scores, private risk budgets, aggregate usage, masked rankings, and proof-carrying receipts. The commitment is not a substitute for encryption; it is a public binding to a private value.

4.3 Homomorphic Encryption, MPC, Garbled Circuits, and ORAM

Fully homomorphic encryption (FHE) evaluates circuits over ciphertexts. Approximate arithmetic schemes such as CKKS are relevant for vector and neural arithmetic, but have cost, precision, and circuit-depth constraints. Secure multi-party computation (MPC) distributes computation across parties such that privacy holds against a bounded collusion threshold. Garbled circuits are often effective for boolean or branching logic. Oblivious RAM (ORAM) hides access patterns at a bandwidth and complexity cost.

cTEEs treat these as operator families:

$$\text{Op} \in \{\text{Public}, \text{Redacted}, \text{FHE}, \text{MPC}, \text{GC}, \text{ORAM}, \text{Guardian}, \text{TEE}\}.$$

The envelope chooses the cheapest operator family that satisfies the sensitivity policy. It does not require every operation to be FHE.

4.4 Receipts

A receipt is a signed or hash-addressed statement that binds:

$$R = H(\text{actor}, \text{operation}, \text{inputs}, \text{outputs}, \text{policy}, \text{authority}, \text{stateRoot}, \text{time}).$$

Receipts let the system prove that an operation occurred under a policy without revealing protected payloads. A receipt may include commitments instead of plaintext outputs.

5 cTEE Syntax

A cTEE scheme is a tuple of algorithms:

$$\Pi_{\text{cTEE}} = (\text{Setup}, \text{Open}, \text{Seal}, \text{Patch}, \text{Compile}, \text{Eval}, \text{Declassify}, \text{Exit}, \text{Commit}, \text{Restore}).$$

Setup(1^λ). Generates system parameters, encryption profiles, commitment parameters, supported operator families, and receipt schemas.

Open(*User*, *P*). Opens a private workspace under policy *P*. The user sees ordinary files and folders; the node sees only permitted views.

Seal(*W*, *M*, *role*). Encrypts a workspace object *M* into a sealed object *C* with role metadata, state roots, and policy-bound AAD.

Patch(*W*, Δ). Converts a user edit into an encrypted patch, a public trunk patch, or a redacted projection update.

Compile(*task*, *P*). Compiles a task into a **PrivateWorkspaceCapsule**: visible context, protected refs, allowed remote ops, forbidden plaintext classes, leakage profile, and required receipts.

Eval(*Node*, *capsule*). Executes public, redacted, masked, encrypted, or guardian-mediated operations and emits commitments and receipts.

Declassify(*Y*, *P*, *grant*). Converts a protected output into a visible result or denies/escalates it.

Exit(*a*, *P*, *grant*). Performs an external action through a capability exit, such as broker order signing, API write, email send, or deployment.

Commit(*R*). Records refs, receipts, commitments, and state roots in STATELOG.

ProveCustody(*trace*, *P*). Produces a **CustodyProof** from the sensitivity manifest, custody derivation, mount graph, receipts, leakage budget, and state roots.

VerifyCustody(π , *P*). Verifies that a custody proof π is consistent with policy, receipts, and state roots.

Restore(*archive*, *P*). Rehydrates workspace state after authority checks, hash verification, decryption, and state-root validation.

6 Leakage-Typed cTEE-IR

The compiler is the enforcement point. A cTEE implementation should not rely on operator authors to remember which data may touch an untrusted node. It should compile agent work into a leakage-typed intermediate representation whose typing rules reject accidental plaintext custody before execution.

6.1 Sensitivity Lattice

Let sensitivity labels form a partial order:

$$\text{public} \sqsubseteq \text{redacted} \sqsubseteq \text{sealed} \sqsubseteq \text{guardian} \sqsubseteq \text{authority}.$$

The order is not a disclosure permission. It is a custody boundary. Public and approved redacted values may be materialized on NODE. Sealed values may appear only as ciphertexts, commitments, shares, or handles. Guardian values may be materialized only inside GUARDIAN or a threshold policy path. Authority values may never be handed to NODE; they can only be exercised through capability exits.

Each expression has a type and label:

$$\Gamma \vdash e : \tau^\ell, \quad \ell \in \{\text{public}, \text{redacted}, \text{sealed}, \text{guardian}, \text{authority}\}.$$

The label of a derived expression is at least the join of its dependencies:

$$\ell(e_1 \circ e_2) \supseteq \ell(e_1) \sqcup \ell(e_2).$$

6.2 Custody Kinds

Sensitivity labels say what a value is. Custody kinds say where its plaintext may ever be materialized:

$$\Gamma \vdash e : \tau^\ell @ \kappa.$$

The base custody kinds are:

Public. Plaintext may be materialized on the node.

Redacted. A policy-approved projection may be materialized on the node, but the source value may not.

Sealed. The node may hold ciphertexts, commitments, shares, or opaque refs only.

GuardianOnly. Plaintext may be materialized only inside the authority view, local client, browser, mobile guardian, or threshold path.

CryptoOperator. The value may participate in an approved FHE, MPC, garbled-circuit, ORAM, local-guardian, or threshold operator.

CapabilityOnly. The value is authority, not data. The node may request an action, but cannot read the credential or grant material.

NeverRemotePlaintext. The value has no valid node-plaintext projection under the current policy.

A remote plan type-checks only when every edge into the node has a custody kind that admits the planned node-visible representation. This makes plaintext custody a property of the compiled graph, not a convention remembered by tool authors.

6.3 Remote Admissibility

An operator op may be scheduled on NODE only if the compiler can derive:

$$\Gamma, P \vdash op(e_1, \dots, e_n) \Downarrow_{\text{Node}} (y, R, \mathcal{L}_{op})$$

subject to one of the following admissibility rules:

1. **Plain remote rule.** Every input label is **public** or policy-approved **redacted**, and the output is not more sensitive than policy permits.
2. **Cryptographic remote rule.** Any sealed input is represented on NODE only as FHE ciphertext, MPC share, garbled-circuit wire label, ORAM handle, encrypted blob ref, or commitment. The operator family must match the representation.
3. **Guardian-mediated rule.** The node may issue a bounded query to GUARDIAN and receive only an approved response class: selection bit, masked score, redacted critique, encrypted result, or declassification denial.
4. **Capability-exit rule.** Authority-labeled values cannot be read. The node may propose an action; GUARDIAN, the authority plane, or the execution boundary decides whether to execute it through a scoped capability exit.

If no rule applies, the compiler must reject the remote plan or repartition the task. This is the practical meaning of no-plaintext custody.

6.4 Receipt-Carrying Types

Remote execution returns values with receipt obligations:

$$\Gamma, P \vdash op : \tau_1^{\ell_1} \times \dots \times \tau_n^{\ell_n} \rightarrow \tau_o^{\ell_o} [\mathcal{R}, \mathcal{L}_{op}].$$

Here $\mathcal{R}$ is the set of required receipts, for example:

- `PrivateInferenceReceipt;`
- `PrivateOperatorReceipt;`
- `CounterfactualLatticeReceipt;`
- `ArtifactRecordedReceipt;`
- `DeclassificationReceipt;`
- `CapabilityExitReceipt.`

A value whose receipt obligations are unsatisfied cannot be committed as admitted workspace truth.

6.5 Compiler Invariant

Invariant 1 (No Plaintext by Construction). *If the cTEE compiler accepts a remote plan for NODE, then every sealed, guardian, or authority dependency in that plan is either absent from the node view, represented by an approved cryptographic carrier, or exercised through an approved capability exit.*

6.6 Proof-Carrying Workspace

A cTEE run can carry a custody proof: a compact verifier-facing object that binds the type derivation, runtime mount, private operators, receipts, leakage budget, and resulting state root.

```
CustodyProof:
  proof_id: custody_proof://...
  workspace_id: workspace://...
  task_id: task://...
  policy_hash: sha256:...
  sensitivity_manifest_hash: sha256:...
  custody_type_derivation_hash: sha256:...
  mount_graph_hash: sha256:...
  remote_admissibility_derivation_hash: sha256:...
  candidate_lattice_commitments:
    - commitment://...
  private_operator_receipts:
    - receipt://...
  declassification_receipts:
    - receipt://...
  capability_exit_receipts:
    - receipt://...
  leakage_receipts:
    - receipt://...
  state_root_before: sha256:...
  state_root_after: sha256:...
  verifier_result:
    no_plaintext_custody | rejected | inconclusive
```

Definition 1 (Proof-Carrying Workspace). *A workspace result is proof-carrying when an independent verifier can check, from the custody proof and referenced receipts, that every protected dependency in the accepted run satisfied a remote admissibility rule or was explicitly declassified by policy.*

The proof does not reveal protected plaintext. It proves the custody route: which values were public, which were redacted, which remained sealed, which entered private operators, which were declassified, and which authority exits were exercised.

7 Core Objects

7.1 PrivateWorkspace

```
PrivateWorkspace:
  workspace_id: workspace://...
  owner_ref: principal://...
  policy_hash: sha256:...
  state_root: sha256:...
  public_trunk_refs: [artifact://...]
  encrypted_object_refs: [artifact://...]
  redacted_projection_refs: [projection://...]
  guardian_ref: guardian://...
  leakage_profile_ref: leakage://...
  receipt_refs: [receipt://...]
```

7.2 PrivateWorkspaceCapsule

```
PrivateWorkspaceCapsule:
  capsule_id: private_workspace_capsule://...
  task_id: task://...
  node_id: runtime://...
  visible_context:
    - public_dataset_ref
    - redacted_summary
    - public_schema
    - bounded_objective
  protected_context:
    - encrypted_payload_ref: artifact://...
    - alpha_seal_ref: alpha_seal://...
    - private_memory_ref: memory://...
  allowed_remote_ops:
    - public_model_inference
    - tensor_compute
    - masked_score_eval
    - encrypted_checkpoint_write
  forbidden_remote_ops:
    - plaintext_secret_read
    - broker_key_read
    - live_portfolio_plaintext_read
    - unbounded_external_action
  leakage_profile:
    visible: [task_type, tensor_dimensions, runtime_cost]
    padded: [output_size, schedule_window]
  receipts:
    required: [private_inference, artifact_recorded, capability_exit_if_action]
```

7.3 AlphaSeal

AlphaSeal is the sealed private-head pattern. It is named for quantitative strategies, but applies to any high-value private decision logic.

```
AlphaSeal:
  alpha_seal_id: alpha_seal://...
  owner_ref: principal://...
  strategy_commitment: sha256:...
  public_trunk_ref: model://... | workflow://...
  private_head:
    representation: local_only | masked_linear | fhe_score |
                    garbled_rules | mpc_share | committed_witness
    encrypted_payload_ref: artifact://...
    leakage_profile_ref: leakage://...
  policy:
    max_notional: ...
    max_daily_loss: ...
    allowed_markets: [...]
    allowed_brokers: [...]
  authority:
    authority_grant_ref: grant://...
    guardian_ref: guardian://...
```

7.4 AutonomyLease

An `AutonomyLease` lets a private workspace node keep working while the user is offline, without granting plaintext custody:

```
AutonomyLease:
  lease_id: autonomy_lease://...
  owner_ref: principal://...
  node_ref: runtime://...
  valid_until: timestamp
  allowed_without_user_online:
    - refresh_public_data
    - run_public_model_inference
    - update_encrypted_state
    - evaluate_alpha_seal
    - propose_order_intent
    - draft_redacted_report
  forbidden_without_step_up:
    - reveal_strategy_source
    - reveal_pii
    - widen_broker_scope
    - exceed_risk_limits
    - export_private_memory
  guardian_policy:
    required: true
    quorum: 1-of-1 | 2-of-3 | policy_defined
```

7.5 NodeMeasurementReceipt

Consumer GPUs such as rented RTX 3090 nodes generally do not provide the same hardware confidential-computing story as newer attested accelerator stacks. Still, a cTEE can use one-time or periodic node measurement for integrity, metering, and reproducibility. Measurement does not make plaintext safe on the node. It says which public runtime was launched and which public artifacts were bound to the run.

```
NodeMeasurementReceipt:
  receipt_id: receipt://...
  node_ref: runtime://...
  boot_epoch: integer
  runtime_manifest_hash: sha256:...
  container_or_vm_image_hash: sha256:...
  driver_profile_hash: sha256:...
  gpu_profile:
    model: RTX_3090 | H100 | ...
    memory_gb: ...
    cuda_profile_hash: sha256:...
  public_model_hashes: [sha256:...]
  public_trunk_root: sha256:...
  forbidden_plaintext_classes: [...]
  leakage_profile_hash: sha256:...
  measurement_method:
    reproducible_boot | provider_attestation | tee_attestation |
    remote_challenge | policy_declared
  signature: ...
```


This receipt helps answer: did the node run the expected public trunk, model, and operator manifest? It does not answer: can the provider see normal plaintext memory? The cTEE privacy answer remains no-plaintext custody.

8 Private Workspace File System

8.1 Encrypted Names and Patches

Filenames can leak alpha. A path such as:

```
/strategies/smallcap-biotech-post-fda-momentum/
```

may reveal more than the file body. A private workspace therefore supports encrypted names:

$$nameId = H(K_{name} \parallel path \parallel epoch),$$

and stores the display name inside an encrypted metadata object. The node may see:

```
/workspace/public_trunk/  
/sealed/alpha_seal://...  
/objects/sha256/3f/...
```

while the user sees normal files and folders.

User edits are represented as encrypted patches:

$$\Delta_C \leftarrow \text{AEAD.Enc}_{K_W}(N, A, \langle objectId, parentHead, patch, editor, policyHash \rangle).$$

The state log commits:

$$head' = H(head \parallel H(\Delta_C) \parallel policyHash \parallel receiptRoot).$$

8.2 Public Trunk and Private Head

Every private workspace object is classified:

$$role(o) \in \{\text{publicTrunk}, \text{redactedStub}, \text{sealedPrivate}, \text{guardianOnly}, \text{actionAuthority}\}.$$

Only `publicTrunk` and approved `redactedStub` objects may be mounted as plaintext on NODE. A public trunk should contain generic pipelines, schemas, backtest engines, public data transforms, report templates, and public model routes. A private head contains alpha formulas, coefficients, thresholds, risk gates, live portfolio, private memory, broker keys, and final action logic.

9 Plaintext-Free Runtime Mounting

Plaintext-Free Runtime Mounting is the runtime interface that makes cTEE useful for ordinary tools, files, shells, model servers, and model calls. A runtime mount is not a decrypted filesystem mount. It is a typed projection of a private workspace into a runtime view:

$$\text{MountView} = \langle S_{\text{pub}}, \text{Redact}(S_{\text{red}}), \text{Ref}(\text{Enc}(S_{\text{seal}})), \text{Handle}(S_{\text{guard}}), \text{Exit}(S_{\text{auth}}) \rangle.$$

Tools and models may consume public content, redacted projections, encrypted refs, candidate commitments, private-head handles, declassification request handles, and capability-exit handles. They must not receive protected files, prompts, memory, strategy source, credentials, or private authority as provider-readable plaintext. Plaintext-Free Model Mounting is the model-facing specialization of this broader runtime mount.

```
PlaintextFreeModelMount:
  mount_id: model_mount://...
  workspace_id: workspace://...
  node_ref: runtime://...
  visible_entries:
    - public_file
    - public_dataset
    - redacted_projection
  protected_entries:
    - encrypted_workspace_object_ref
    - private_head_handle
    - declassification_request_handle
    - capability_exit_handle
  forbidden_plaintext_classes:
    - pii
    - strategy_source
    - private_memory
    - broker_secret
    - live_portfolio
  leakage_profile_ref: leakage://...
  deterrence_detection_profile_ref: deterrence://...
  receipts_required:
    - ModelMountReceipt
    - PrivateInferenceReceipt
    - LeakageReceipt
```

This is the precise role of ciphering in cTEEs. Ciphering is useful only when the node never receives the deciphering key and never materializes protected plaintext or plaintext-equivalent private activations. The model mount prevents accidental custody: the remote model can reason over what it is allowed to see, and must ask private handles for anything else.

Runtime-mount doctrine. Tools and models may operate against the workspace. The node must not become the plaintext custody domain for protected workspace state.

10 Cryptographic Operator Plane

Plaintext-Free Runtime Mounting prevents accidental plaintext custody. The Cryptographic Operator Plane handles the cases where the node must participate in a protected subcomputation without learning the protected inputs, private policy, or final authority.

The default routing rule is:

public/generic work	-> untrusted rented GPU
private file view	-> guardian/client
private head/scoring	-> FHE/MPC/local/threshold operator
private retrieval	-> ORAM/local/private index

```
external action      -> capability exit
state truth          -> StateLog
```

This does not change the token speed of the main public/generic LLM path. The GPU still performs ordinary inference over public trunks, redacted projections, public data, public tools, candidate lattices, and generic simulation. Overhead appears only at protected boundaries: private scoring, private selection, private retrieval, declassification, and action authorization. When the compiler chooses counterfactual lattice expansion, the per-token kernels remain ordinary plaintext GPU kernels, but the run may spend more public tokens to reduce private-choice leakage.

Operator-plane doctrine. cTEE does not make hostile hardware safe for plaintext. It replaces the need for hardware secrecy when protected work can be routed through no-plaintext-custody flows, guardian evaluation, or cryptographic operators.

10.1 Default Two-Party Topology

MPC and threshold operators require more than one logical party. In a default cTEE deployment, the additional logical party is the user's existing authority surface:

Party A:

untrusted DePIN / rented GPU node

Party B:

authenticated browser
local desktop harness when available
mobile guardian
CLI signer
authority-plane policy/key path
enterprise key service when org-managed

The user should not need to rent or manage additional GPU nodes for the default privacy path. Managed non-colluding committees remain optional escalation paths for higher-assurance, unattended, or enterprise workflows. FHE-only operators may run on a single untrusted node when the node receives only ciphertexts and evaluation keys, never decryption keys.

Second-party doctrine. The second logical party is the authenticated authority surface by default. Managed committees are optional escalation paths, not the ordinary user mental model.

10.2 Policy Object

```
CryptographicOperatorPolicy:
  policy_id: crypto_op_policy://...
  workspace_id: workspace://...
  protected_classes:
    - pii
    - strategy_source
    - private_memory
    - broker_secret
    - live_portfolio
  allowed_operator_families:
```

```

- fhe_linear
- fhe_approx
- mpc_nonlinear
- garbled_boolean
- oram_lookup
- local_guardian
- threshold_guardian
default_second_party:
  authority_surface
second_party_refs:
- browser_session://...
- mobile_guardian://...
- cli_signer://...
- authority_plane://...
fallback_order:
- local_guardian
- threshold_guardian
- fhe_linear
- mpc_nonlinear
- deny_or_escalate
max_latency_budget_ms: ...
leakage_budget_ref: leakage://...
receipts_required:
- PrivateOperatorReceipt
- LeakageReceipt

```

10.3 Private Operator Receipt

```

PrivateOperatorReceipt:
  receipt_id: receipt://...
  policy_ref: crypto_op_policy://...
  run_id: run://...
  capsule_id: private_workspace_capsule://...
  operator_family:
    fhe_linear | fhe_approx | mpc_nonlinear |
    garbled_boolean | oram_lookup |
    local_guardian | threshold_guardian
  node_ref: runtime://...
  second_party_ref:
    browser_session://... | mobile_guardian://... |
    cli_signer://... | authority_plane://... |
    threshold_guardian://...
  protected_input_commitments:
    - commitment://...
  public_input_refs:
    - artifact://...
  output_commitment: commitment://...
  leakage_profile_ref: leakage://...
  policy_hash: sha256:...
  plaintext_sensitive_classes_on_node:
    - none
  status:
    success | failure | denied | escalated

```

10.4 Confidential-Compute Replacement Boundary

Proposition 1 (Hardwareless Confidential-Compute Replacement Boundary). *A cTEE deployment replaces hardware confidential computing for a workload when every protected dependency in that workload is either absent from the node view, represented by an approved cryptographic carrier, evaluated by the guardian or threshold path, or exercised through a capability exit, and all transitions are bound by receipts and leakage policy.*

This is a workload-level claim, not a universal machine-level claim. It is stronger than trusting a rented host with plaintext, but narrower than saying arbitrary private plaintext computation runs at normal speed on hostile hardware.

11 Deterrence and Detection

Deterrence and detection are not privacy primitives. They are attribution, abuse-response, and dispute primitives. They matter because rented nodes, marketplace workers, and service providers can still deny service, replay work, copy public/redacted outputs, or attempt to lure users into unsafe plaintext mounts.

A high-value cTEE workspace may bind every capsule and model mount to a deterrence profile:

```
DeterrenceDetectionProfile:
  profile_id: deterrence://...
  provider_bound_nonce: sha256:...
  node_bound_watermark: sha256:...
  canaries:
    - decoy_strategy
    - decoy_credential
    - synthetic_pii
    - watermark_phrase
    - honeypoint_endpoint
  watermarking:
    candidate_lattice_watermark: true
    report_text_watermark: true
    code_patch_watermark: true
  monitoring:
    scan_public_leaks: true
    broker_honeypoint_alerts: true
    suspicious_replay_detection: true
```

Canaries must be synthetic. They must not be real secrets, and they must be labeled so they do not contaminate training, memory, settlement, or user decisions. Watermarks must identify a capsule, provider, node, or run without revealing private strategy content.

```
DeterrenceDetectionReceipt:
  receipt_id: receipt://...
  profile_ref: deterrence://...
  event_type:
    canary_planted | watermark_bound | honeypoint_bound |
    canary_tripped | suspicious_replay_detected | leak_scan_completed
  bound_refs:
    - model_mount://...
    - private_workspace_capsule://...
```

```

public_evidence_refs:
  - artifact://...
private_evidence_commitments:
  - commitment://...
action:
  none | warn_user | revoke_node | rotate_keys |
  open_dispute | slash_provider | quarantine_workspace

```

This layer makes cTEE better suited to marketplaces such as service outcome networks and worker catalogs: a dispute can show that a provider-bound canary, watermark, or replay signature escaped without requiring the user to reveal the real private workspace.

12 Protocols

Protocol. Remote Node Enrollment

1. NODE publishes hardware profile, runtime manifest, driver profile, public model hashes, supported operator families, and storage policy.
2. GUARDIAN or the coordinator challenges NODE to produce a `NodeMeasurementReceipt`.
3. STATELOG records the measurement receipt, node policy, revocation epoch, and accepted leakage profile.
4. The scheduler may route public trunk work to NODE only while the measurement and lease remain valid.
5. Protected workspace classes remain forbidden regardless of measurement quality unless a stronger policy explicitly enables a hardware TEE mode.

Protocol. Open Private Workspace

1. USER selects a runtime node and a privacy policy P .
2. GUARDIAN derives or authorizes workspace keys K_W , name keys K_{name} , and declassification policy D .
3. STATELOG creates a workspace genesis operation:

$$G = H(\text{workspaceId}, \text{owner}, \text{policyHash}, \text{guardianRef}, \text{schemaVersion}).$$

4. BLOBSTORE receives encrypted root objects and sealed archive refs.
5. NODE receives the public trunk, redacted projections, encrypted refs, and allowed operator manifest.
6. The user sees a normal workspace view. The node sees no protected file plaintext by default.

Protocol. Compile Private Workspace Capsule

1. Classify task inputs by sensitivity class.
2. Build visible context from public trunk and redacted projections.
3. Replace protected references with encrypted artifact refs, commitments, sealed private heads, or guardian calls.
4. Attach forbidden plaintext classes and allowed operator families.
5. Attach a leakage profile $\mathcal{L}$ and receipt requirements.
6. Commit the capsule hash and policy hash in STATELOG.

Protocol. Declassification Gate

1. Protected output Y_C or commitment C_Y is produced.
2. NODE submits a `PrivateInferenceReceipt`.
3. GUARDIAN checks policy, autonomy lease, revocation epoch, leakage profile, and requested disclosure class.
4. If allowed, GUARDIAN decrypts or reconstructs Y , scans it, and emits a declassification receipt.
5. If the output implies an external action, the action is routed through a `CapabilityExit`. The node never receives durable raw credentials.

13 Security Definitions

13.1 No-Plaintext-Custody

Definition 2 (No-Plaintext-Custody). *A cTEE scheme satisfies no-plaintext-custody (NPC) for protected class set $\mathcal{C}$ and leakage function $\mathcal{L}$ if for every probabilistic polynomial-time adversary $\mathcal{A}$ controlling NODE, there exists a simulator $\mathcal{S}$ such that for any two workspaces W_0, W_1 with $\mathcal{L}(W_0) = \mathcal{L}(W_1)$, the adversary’s view during node execution is computationally indistinguishable from $\mathcal{S}(\mathcal{L}(W_b))$, excluding values explicitly declassified by policy.*

$$|\Pr[\mathcal{A}(\text{View}_{\text{Node}}(W_0)) = 1] - \Pr[\mathcal{A}(\text{View}_{\text{Node}}(W_1)) = 1]| \leq \text{negl}(\lambda).$$

NPC is not the same as hiding all metadata. It hides protected plaintext modulo declared leakage and explicit declassification.

13.2 Authority Safety

Definition 3 (Authority Safety). *A cTEE satisfies authority safety if a node-controlled adversary cannot produce an accepted external effect a unless there exists an unexpired authority grant g , a policy P , and a valid receipt chain proving:*

$$\text{allowed}(P, g, a) = \text{true}.$$

External effects include broker orders, payments, email sends, API writes, deployments, secret exports, and persistent state imports.

13.3 Receipt Binding

Definition 4 (Receipt Binding). *A receipt R is binding if no efficient adversary can produce two accepted interpretations of the same receipt hash with different actor, policy, authority, input commitment, output commitment, or resulting state root.*

13.4 Restore Validity

Definition 5 (Restore Validity). *A restored workspace state is valid if the restore operation verifies archive hashes, decrypts under authorized policy, checks object heads, checks state root, records import receipts, and rebuilds projections from accepted refs rather than silently mutating local files.*

14 Security Theorems

Assumption 1 (Primitive Security). *AEAD is IND-CCA and INT-CTXT secure; commitments are binding and hiding; signatures are existentially unforgeable; hash functions are collision resistant; guardian thresholds satisfy their stated corruption bounds; and FHE/MPC/GC/ORAM operators satisfy their standard security definitions when used.*

Assumption 2 (No Accidental Plaintext). *The implementation enforces the capsule policy: protected classes are not mounted, logged, prompted, cached, traced, or sent to NODE as normal plaintext unless a declassification receipt explicitly allows it.*

Theorem 1 (NPC Security of Base cTEE). *Under the primitive security and no-accidental-plaintext assumptions, the base cTEE construction satisfies no-plaintext-custody for protected workspace objects whose plaintext never leaves USER or GUARDIAN.*

Proof sketch. The node receives public trunk objects, redacted projections, encrypted payloads, commitments, receipts, and declared leakage. By AEAD security, sealed workspace objects are indistinguishable from random strings to a node lacking keys. By commitment hiding, committed private values reveal only declared commitment metadata. By the no-accidental-plaintext assumption, no alternate path gives the node protected plaintext through logs, prompts, mounts, caches, or traces. Therefore the node view can be simulated from public trunks, redacted projections, ciphertext lengths, commitments, receipts, and leakage $\mathcal{L}$. $\square$

Theorem 2 (Custody Type Soundness). *If a remote plan type-checks under the custody-kind rules, all required receipts verify, and the runtime enforces the checked mount graph, then no value typed GuardianOnly, CapabilityOnly, or NeverRemotePlaintext is materialized as node-observable plaintext. Values typed Sealed or CryptoOperator appear in the node view only through their admitted ciphertext, share, commitment, handle, or operator transcript representations.*

Proof sketch. Proceed by induction over the accepted remote execution derivation. Source values enter the node only through mount edges. The plain remote rule admits only public or approved redacted projections. The cryptographic remote rule admits only carrier representations whose security is delegated to the underlying primitive. The guardian-mediated rule admits only the approved response class and receipt. The capability-exit rule admits an action request but not the authority value. Since every derived edge must satisfy one of these rules and the runtime enforces the checked mount graph, no forbidden custody kind has a derivation into node plaintext. $\square$

Corollary 1 (Proof-Carrying NPC). *If a **CustodyProof** verifies for a run, and its referenced receipt and state-root commitments are binding, then the accepted result carries checkable evidence that protected workspace values did not enter node plaintext custody except through explicit declassification receipts.*

Theorem 3 (Authority Safety of Capability Exits). *If signature unforgeability holds and the authority verifier accepts only unexpired grants satisfying policy predicates, then a node adversary cannot cause an accepted external effect outside the granted scope except with negligible probability.*

Proof sketch. Every external effect requires a capability-exit receipt bound to a grant, policy hash, action summary, input commitment, output commitment, and signature or threshold approval. Producing an accepted effect outside policy requires either forging an authority signature, finding a binding collision in the receipt, or exploiting a verifier bug outside the model. $\square$

Proposition 2 (No Free Full-Speed Private LLM Inference). *On a root-controlled consumer GPU without trusted hardware, trusted guardian participation, or cryptographic private evaluation, arbitrary private LLM inference over protected plaintext at ordinary plaintext token speed cannot be claimed by a software wrapper alone.*

Argument. If the node performs the same plaintext matrix and attention operations as a normal LLM over private tokens or embeddings, then intermediate plaintext or semantically equivalent representations must exist in node-observable process memory, logs, IPC, kernel buffers, driver buffers, or GPU-visible memory under the adversary model. A software wrapper that leaves those representations on the node changes neither the information flow nor the provider’s root access. To remove the plaintext from the node, the computation must move to another trust domain, use hardware isolation, or use cryptographic representations with nonzero overhead. Thus a pure software wrapper cannot honestly claim arbitrary private LLM inference at normal plaintext speed on the adversarial node. $\square$

15 Private Agency Transform

The negative result above rules out magic software wrapping for arbitrary private LLM inference. The positive result is that many useful agent tasks do not need arbitrary private inference. They need private agency: choosing, ranking, rejecting, authorizing, or declassifying among candidate traces.

15.1 Candidate-Selection Reducibility

Definition 6 (Candidate-Selection-Reducible Task). *A private agent task T with public projection x_{pub} , private state s , action space $\mathcal{A}$, and private utility $U_s : \mathcal{A} \rightarrow \mathbb{R}$ is (K, ϵ, ρ) -candidate-selection-reducible if there exists a public generator G such that:*

$$C_K \leftarrow G(x_{\text{pub}})$$

produces K candidates and, with probability at least ρ , contains an action $a \in C_K$ whose private utility is within ϵ of the best available action:

$$U_s(a) \geq \max_{a' \in \mathcal{A}} U_s(a') - \epsilon.$$

For such tasks, the expensive language-model work is public proposal generation. The private work is selection, validation, reranking, declassification, or authorization over the candidate set.

15.2 Transform

The Private Agency Transform rewrites a secret-bearing agent step:

$$\text{LLM}(x_{\text{pub}}, s) \rightarrow a$$

into:

$$G_{\text{Node}}(x_{\text{pub}}) \rightarrow C_K, \quad H_{\text{Guardian}}(C_K, s, P) \rightarrow a, R, \mathcal{L}.$$

G_{Node} runs at ordinary node-side inference speed over public or redacted context. H_{Guardian} is a private selector implemented by the authority view, a local head, FHE, MPC, garbled circuits, ORAM-backed lookup, or threshold policy. The result is committed through receipts and leakage accounting.

Protocol. Private Agency Transform

1. Compile protected context into custody-typed refs, redacted projections, sealed heads, and private-operator handles.
2. Ask NODE to generate a candidate trace lattice under public or redacted context only.
3. Commit the candidate lattice before private selection.
4. Evaluate private utility, constraints, policy, or authority over the candidate lattice inside the authority surface or cryptographic operator plane.
5. Return only an admitted response class: selected branch, top- m set, masked score, denial, redacted critique, or declassification request.
6. Emit custody, private-operator, leakage, and capability receipts.

15.3 Private Agency Theorem

Theorem 4 (Public-Kernel-Speed Private Agency for Reducible Tasks). *Let T be a (K, ϵ, ρ) -candidate-selection-reducible task. Suppose the public generator G runs on NODE without protected plaintext, the private selector H satisfies the custody-kind rules, and the accepted run emits verifying custody and leakage receipts. Then cTEE can execute T with:*

1. *node-side token throughput equal to the public inference throughput of $G(x_{\text{pub}})$, while total token volume scales with the chosen candidate width, depth, and counterfactual schedule;*
2. *private computation cost proportional to $K \cdot \text{cost}(H)$, not to the full transformer token path;*
3. *an ϵ -optimal action with probability at least ρ , modulo verifier error and declassification policy; and*
4. *no-plaintext-custody for protected state, with node leakage bounded by public projection, candidate lattice, timing/size/schedule observables, and declared selection leakage.*

Proof sketch. Candidate-selection reducibility gives the coverage claim: with probability ρ , G proposes a candidate within ϵ of the private optimum. The node computes only $G(x_{\text{pub}})$, so its token path is the public inference path. The selector H evaluates private utility only over the finite candidate set and therefore pays per-candidate private cost rather than per-token transformer cost. Custody type soundness and proof-carrying NPC give the no-plaintext-custody claim. The remaining node view consists of the public projection, candidate lattice, commitments, receipts, and declared leakage. $\square$

15.4 Candidate Coverage Frontier

The previous theorem hides the most important parameter inside ρ . A more useful formulation exposes candidate redundancy.

Definition 7 (Private-Good Set). *For private state s , policy P , tolerance ϵ , and public action space $\mathcal{A}$, define the private-good set:*

$$\mathcal{A}_\epsilon(s, P) = \{a \in \mathcal{A} : U_s(a) \geq \max_{a' \in \mathcal{A}} U_s(a') - \epsilon \wedge \text{allowed}(P, a)\}.$$

Definition 8 (Candidate Redundancy Mass). *Let $D_G(x_{\text{pub}})$ be the distribution over complete candidate traces emitted by a public generator G . The ϵ -redundancy mass of a task instance is:*

$$r_\epsilon(x_{\text{pub}}, s, P) = \Pr_{a \sim D_G(x_{\text{pub}})}[a \in \mathcal{A}_\epsilon(s, P)].$$

A task family is (ϵ, r) -proposal-redundant when $r_\epsilon(x_{\text{pub}}, s, P) \geq r$ for all states and policies in the family.

Redundancy mass is the amount of useful private choice already present in the public generator's distribution. A high-redundancy task has many public candidates that the private head could safely accept. A low-redundancy task requires private generation, stronger private operators, or a different public abstraction.

Theorem 5 (Coverage-Cost-Leakage Frontier). *For an (ϵ, r) -proposal-redundant task family, sample m complete candidate traces independently from $G(x_{\text{pub}})$, commit the trace set before private selection, and select privately inside the authority view or cryptographic operator plane. Then:*

$$\Pr[\exists a_i \in \mathcal{A}_\epsilon(s, P)] \geq 1 - (1 - r)^m.$$

Therefore, to reach coverage probability ρ , it suffices to choose:

$$m \geq \left\lceil \frac{\ln(1 - \rho)}{\ln(1 - r)} \right\rceil \leq \left\lceil \frac{1}{r} \ln \frac{1}{1 - \rho} \right\rceil.$$

Before declassification, online selection leakage is zero when the node receives no private feedback before the trace-set commitment. Public cost scales with m public traces; private cost scales with m private scores; neither cost scales with a full private transformer execution over the protected state.

Proof sketch. Each complete trace independently lands in the private-good set with probability at least r . The probability that no trace lands in the set is at most $(1 - r)^m$. Taking the complement gives the coverage bound. Solving $1 - (1 - r)^m \geq \rho$ gives the exact expression for m . The upper bound uses $\ln(1 - r) \leq -r$. Zero online selection leakage follows because the node's transcript up to commitment is generated before private selection feedback exists. $\square$

Corollary 2 (Counterfactual Path Privacy Without Tree Explosion). *For a depth- D agent episode, hiding every private branch by naively expanding a K -ary tree may cost K^D public continuations. If complete public traces have redundancy mass r_D , counterfactual trace sampling reaches coverage ρ with:*

$$m = O\left(\frac{1}{r_D} \log \frac{1}{1-\rho}\right)$$

complete traces. Thus path privacy depends on the probability mass of acceptable complete traces, not on the syntactic size of the full branch tree. When r_D is non-negligible, zero online selection leakage is achievable without tree explosion.

Corollary 3 (Redundancy Phase Transition). *For a sequence of depth- D tasks with coverage target ρ :*

1. *If $r_D \geq c > 0$, then the number of complete public traces required for zero online selection leakage is*

$$m = O\left(\log \frac{1}{1-\rho}\right),$$

independent of D and independent of the syntactic branch factor K .

2. *If $r_D = e^{-\alpha D}$ for $\alpha > 0$, then*

$$m = \Theta\left(e^{\alpha D} \log \frac{1}{1-\rho}\right).$$

In this regime, CLPD/CLE no longer gives practical private agency at public kernel speed; the runtime should route to private generation, cryptographic operators, trusted/private compute, or explicit provider trust.

Proof sketch. Substitute the lower bound on r_D into the coverage frontier. Constant redundancy mass gives a trace budget depending only on the desired failure probability. Exponentially decaying redundancy mass makes the inverse-mass term exponential in depth. $\square$

Coverage-frontier doctrine. cTEE’s surprising trade is not “more tokens means more privacy.” It is that public overgeneration buys private-choice privacy in proportion to proposal redundancy. The phase transition is sharp: if acceptable complete traces retain non-negligible probability mass, private-choice privacy costs only bounded public overgeneration; if that mass decays exponentially with task depth, cTEE must stop using CLPD/CLE as the privacy story and route to a stronger private path.

Private-agency doctrine. The surprising positive result is not free private inference. It is that a broad class of agent work can be compiled into normal-kernel public proposal plus small private choice. Online CLPD optimizes for cost; counterfactual CLE spends additional public tokens to buy lower private-choice leakage.

16 Counterfactual Lattice Execution

The Private Agency Transform still has a leakage problem if the private selector feeds choices back to the node at every step. A selected branch among K candidates leaks at most $\log_2 K$ explicit choice bits before side information. Over long agent runs, this can accumulate.

Counterfactual Lattice Execution (CLE) is the high-assurance schedule for CLPD. Instead of telling the node which branch the private head likes and asking it to continue that branch, the node expands many plausible branches. The private path is selected later, inside the authority view or cryptographic operator plane. The node computes counterfactual futures that may never be used.

CLE realizes the coverage-cost-leakage frontier by choosing a public trace budget from the estimated redundancy mass, coverage target, leakage budget, and latency budget.

16.1 Counterfactual Lattice

A counterfactual lattice is a committed directed acyclic graph:

$$\mathcal{G}_{K,D} = (V, E)$$

generated by a public model from public or redacted context, with branching budget K , depth budget D , model hash, policy hash, and nonce. Every node $v \in V$ is a candidate prefix, action, plan fragment, tool proposal, report fragment, trade narrative, patch sketch, or other public candidate artifact. The untrusted node commits to the lattice before private selection:

$$C_{\mathcal{G}} = H(\mathcal{G}_{K,D} \parallel \text{modelHash} \parallel \text{policyHash} \parallel \text{nonce}).$$

16.2 Protocol

Protocol. Counterfactual Lattice Execution

1. The compiler chooses a lattice policy: width, depth, stopping rule, deduplication rule, padding rule, and maximum public-token budget.
2. NODE expands the public lattice without private feedback.
3. NODE commits to the lattice and emits a `CounterfactualLatticeReceipt`.
4. GUARDIAN or the cryptographic operator plane evaluates private utility over the committed lattice.
5. The selected path, top- m set, masked score, denial, or final declassification result is emitted according to policy.
6. The state log records the lattice commitment, custody proof, leakage receipt, private-operator receipts, and any declassification receipts.

`CounterfactualLatticeReceipt:`

```
receipt_id: receipt://...
capsule_id: private_workspace_capsule://...
lattice_commitment: commitment://...
model_hash: sha256:...
```

```

policy_hash: sha256:...
width_budget_k: integer
depth_budget_d: integer
public_token_budget: integer
generation_rule_hash: sha256:...
dedupe_rule_hash: sha256:...
padding_rule_hash: sha256:...
node_ref: runtime://...
state_root: sha256:...

```

16.3 Zero Online Selection Leakage

Theorem 6 (Zero Online Selection Leakage). *In a non-adaptive CLE round, if the lattice generation algorithm receives only public or policy-redacted inputs and the private selector sends no branch choice or private score back to the node before lattice commitment, then the node’s pre-declassification view is independent of the selected private path, except for public lattice parameters, timing, size, schedule, and other declared metadata leakage.*

Proof sketch. The node’s pre-declassification transcript consists of the public projection, generation policy, model/runtime metadata, generated lattice, commitments, receipts, timing, and resource usage. None of these depend on the private selector’s chosen path, because generation completed before private selection feedback. Therefore two private states that choose different paths through the same public lattice induce the same node transcript until an explicit declassification event. $\square$

Corollary 4 (Selection Leakage Compression). *For an online CLPD run with T private branch decisions over K -way choices, explicit selection leakage can scale as $O(T \log K)$. For a non-adaptive CLE round over the same candidate space, explicit online selection leakage is compressed to zero before declassification; remaining leakage is the public lattice itself, side channels, and whatever final output policy reveals.*

CLE does not make compute free. It trades extra public token volume for lower private-choice leakage. This is the core cTEE performance bargain:

privacy by counterfactual public overgeneration, not plaintext trust.

Counterfactual-agency doctrine. The rented GPU can be useful without knowing which future matters. It expands a space of plausible futures; private agency chooses a hidden path through that space.

17 Candidate-Lattice Private Decoding

Candidate-Lattice Private Decoding (CLPD) is the default implementation of the Private Agency Transform for language-model-heavy work. The node produces useful public reasoning candidates, while private state selects, filters, reranks, or authorizes them.

CLPD has two internal schedules:

1. **Online CLPD.** The private selector returns bounded feedback after each candidate expansion. This is cheaper but leaks branch-selection observables.

2. **Counterfactual CLPD.** The node expands a committed lattice before private selection. This costs more public tokens but hides the selected path until declassification.

The user should not choose between these schedules. The compiler selects the schedule from sensitivity, leakage budget, latency budget, and task risk.

17.1 Construction

Let M_{pub} be a public model running on NODE. Let h_{priv} be a private scoring head held by GUARDIAN, MPC, FHE, or a local client. At step t :

1. NODE expands a candidate lattice:

$$\mathcal{Y}_t = \{y_{t:t+B}^{(1)}, \dots, y_{t:t+B}^{(K)}\} \leftarrow M_{\text{pub}}(x_{\text{pub}}, y_{<t}).$$

2. NODE commits to the lattice:

$$C_t = H(\mathcal{Y}_t \parallel \text{modelHash} \parallel \text{policyHash} \parallel \text{nonce}).$$

3. GUARDIAN evaluates private utility:

$$s_i = h_{\text{priv}}(y_{t:t+B}^{(i)}, S_{\text{priv}}, P).$$

4. GUARDIAN returns either an encrypted selection, a masked reranking, a redacted critique, a bounded next prefix, or no online feedback when the run uses the counterfactual schedule. The node receives no private state.
5. Receipts bind C_t , the selected branch commitment, leakage profile, and declassification decision.

17.2 Why It Matters

CLPD preserves the high-throughput path for public transformer work. The GPU does what it is good at: generate candidate continuations, embeddings, public summaries, report drafts, code skeletons, or strategy simulations. The private head does what must remain private: choose among candidates using PII, private policy, live portfolio, alpha formula, or user secrets.

The user-facing product should not expose CLPD as a selectable mode. The user opens a private workspace. The runtime uses CLPD by default wherever protected agency is required and the public/generic candidate generation can be separated from private selection.

CLPD does not implement arbitrary private inference. It implements the default useful class of private agency:

$$\text{public generation} + \text{private selection} + \text{receipted action}.$$

17.3 Security

If the candidate lattice is generated from public or redacted context and the private head reveals only an admitted response class under a leakage profile, then the node learns at most the public candidates, timing, output length, and the declared selection leakage. Online CLPD uses padding, delayed reveal, top- m selection, batching, and decoy selections to reduce selection leakage. Counterfactual CLPD eliminates online branch feedback for the lattice round and charges the remaining disclosure to the final declassification policy.

17.4 Selection Leakage Accounting

CLPD should attach a leakage account to every private selection. A single chosen branch from K candidates leaks at most:

$$L_{\text{select}} \leq \log_2 K$$

bits about the private preference, before considering side information. These bounds account for explicit selection observables, not all semantic inference an adversary may derive from the public candidate set, timing, prior runs, or domain-specific side information. Revealing an unordered top- m set leaks at most:

$$L_{\text{top}m} \leq \log_2 \binom{K}{m}.$$

Revealing a total ranking leaks at most:

$$L_{\text{rank}} \leq \log_2(K!).$$

The receipt should also account for timing, size, and schedule leakage. For a bucketed observable X with b_X possible buckets, use the conservative bound:

$$L_X \leq \log_2 b_X.$$

A practical `LeakageBudget` can therefore be represented as:

```
LeakageBudget:
  budget_id: leakage://...
  task_class: quant_backtest | private_email | legal_review | ...
  candidate_count_k: integer
  reveal_policy: selected_one | top_m | masked_score | denial_only
  selection_bits_bound: number
  timing_bucket_bits_bound: number
  size_bucket_bits_bound: number
  schedule_bucket_bits_bound: number
  cumulative_budget_bits: number
  mitigations:
    - fixed_k
    - top_m_or_decoy_reveal
    - delayed_commit_reveal
    - batch_padding
    - schedule_jitter
    - neutral_labels
```

This is intentionally conservative. It does not prove semantic secrecy by itself; it makes leakage explicit enough for product policy. A high-value quant workspace may set a low daily leakage budget, require decoy reveals, and stop or escalate once cumulative budget is exhausted.

17.5 Agent Use Cases

- **Quant.** The node generates candidate trade narratives or factor sets; the private alpha head scores them with sealed coefficients and live portfolio constraints.

- **Legal.** The node drafts public clause alternatives; a private policy head selects clauses compatible with confidential negotiation strategy.
- **Personal assistant.** The node drafts generic responses; the guardian filters against private user history and declassifies only approved text.
- **Code agent.** The node proposes public trunk patches; private workspace objects stay local, and sensitive diffs are applied through encrypted patches.

18 Default cTEE Agent Execution

A private agent harness using cTEE should route every operation by sensitivity, with CLPD as the default protected-agency path:

Operation	Allowed on node	Protected path
Model mount	Public/redacted view and private handles	ModelMountReceipt
Public LLM inference	Plaintext public prompt	None required
Model API call	Public/redacted/declassified input	Posture disclosure
Redacted summary	Redacted projection	Redaction receipt
Private file edit	Encrypted patch only	Browser/device/guardian view
Private retrieval	Encrypted refs or redacted snippets	Local/guardian/ORAM path
Protected agency	Candidate lattice on node	AlphaSeal/guardian selection
CF lattice	Committed public lattice	Counterfactual lattice receipt
Private operator	Ciphertext, share, commitment, or handle	PrivateOperatorReceipt
Private scoring	Masked, FHE, MPC, GC, or committed witness	Guardian/private head
External action	Proposal only	Capability exit
Restore/import	Encrypted archive fetch	Authority and state-root checks

19 Quantitative Strategy Pattern

For the reference quant workflow, the private workspace separates:

$$\text{Strategy} = \text{PublicTrunk} \parallel \text{AlphaSeal}.$$

PublicTrunk may include data loaders, generic factor libraries, backtest engines, public market data transforms, report templates, simulation kernels, and public model routes. **AlphaSeal** contains signal formulas, private coefficients, thresholds, risk gates, live portfolio constraints, broker keys, and final order logic.

The rented GPU can run:

- market data ingestion;
- public feature tensor generation;
- public model inference;
- public or redacted research;
- public backtests;

- encrypted checkpointing;
- candidate ranking under masked or committed heads;
- report drafting over redacted observations.

The rented GPU must not receive:

- strategy source plaintext;
- private coefficients and thresholds;
- broker credentials;
- live portfolio plaintext;
- final order logic;
- unrestricted signing authority.

19.1 End-to-End Quant Walkthrough

A private quant workspace can still feel like normal remote development:

1. The user opens `signals.py` in the browser, local agent harness, or CLI authority view. The plaintext file is decrypted there, not in the rented node's workspace.
2. The edit is saved as an encrypted patch with neutral object identifiers and committed to the state log with policy, parent head, and receipt refs.
3. The rented GPU node receives the public trunk, public market-data tensors, redacted projections, encrypted refs, and the allowed operator manifest.
4. The node runs public backtests, feature generation, model inference, and candidate trade or report generation at ordinary GPU speed.
5. **AlphaSeal** scores, filters, or reranks candidate trades through the guardian, threshold path, masked head, or cryptographic operator. The node sees only the allowed response class and leakage receipt.
6. The authority plane signs, denies, or escalates an order intent through a configured capability exit. Broker keys and unrestricted order logic are never handed to the node.
7. The state log records the encrypted patch, artifact refs, candidate commitment, leakage receipt, declassification receipt, and capability-exit receipt.

The product promise is simple: the user edits and runs a private workspace; the provider sees compute jobs, public trunks, ciphertexts, commitments, receipts, and bounded disclosures.

20 Leakage Profiles

A cTEE cannot hide everything. It must declare leakage:

$$\mathcal{L} = \langle \text{taskType}, \text{modelFamily}, \text{publicDataset}, \text{tensorShape}, \\ \text{jobDuration}, \text{outputLength}, \text{schedulePattern}, \\ \text{storageSize}, \text{receiptTypes} \rangle.$$

Mitigations include:

- neutral names for paths and jobs;
- encrypted file and folder names;
- batching and schedule jitter;
- padded output sizes;
- decoy jobs for high-value workflows;
- redacted trace and prompt policies;
- ORAM or private lookup for access-pattern-sensitive workflows.

The leakage profile is a first-class object because it is the line between a credible privacy claim and marketing fog.

20.1 Leakage Receipts

Every CLPD selection should emit a leakage receipt. An implementation may also include the same leakage fields in the `PrivateInferenceReceipt`:

```
LeakageReceipt:
  receipt_id: receipt://...
  capsule_id: private_workspace_capsule://...
  candidate_lattice_commitment: commitment://...
  leakage_profile_ref: leakage://...
  selection_policy: selected_one | top_m | denial_only | masked_score
  selection_bits_bound: number
  timing_bucket_bits_bound: number
  size_bucket_bits_bound: number
  cumulative_budget_before: number
  cumulative_budget_after: number
  mitigation_refs:
    - padding_policy://...
    - schedule_jitter_policy://...
    - decoy_policy://...
```

The purpose is operational: a user, service contract, or verifier can decide whether continued private work is still within the promised leakage envelope.

21 Implementation Guidance

21.1 Harness Integration

Any agent harness can implement cTEE:

- route work by sensitivity before tool/model execution;
- store private workspace objects as encrypted refs;
- send only public trunks, redacted projections, commitments, or sealed capsules to untrusted nodes;
- bind every protected output to a receipt;
- route external effects through capability exits;
- make declassification explicit in UI and logs;
- make unsafe plaintext mounts visible and hard to choose.

The concept is not tied to one product. A private agent harness, a marketplace worker runtime, an enterprise runner, or an open-source autonomous-code agent can adopt the same envelope.

21.2 Minimum Viable cTEE

The smallest useful implementation has:

1. encrypted workspace object store with AEAD and state-root binding;
2. encrypted or neutral filenames for sensitive paths;
3. public trunk/private head classification;
4. Plaintext-Free Runtime Mounting with `ModelMountReceipt` for model views;
5. private workspace capsule compiler;
6. Candidate-Lattice Private Decoding for protected agency by default;
7. Counterfactual Lattice Execution for high-sensitivity protected-agency rounds where online branch feedback would exceed leakage budget;
8. Cryptographic Operator Plane with operator policy and private-operator receipts;
9. guardian or client-side declassification;
10. capability exit for secrets and external actions;
11. synthetic canaries or provider-bound watermarks for high-value workspaces, recorded through deterrence/detection receipts;
12. receipt log with private inference, declassification, artifact, and action receipts.

21.3 Conformance Tests

1. A protected file never appears as plaintext on NODE during normal open, edit, run, checkpoint, trace, or restore.
2. A protected filename does not appear in node-visible paths in high security mode.
3. A model prompt containing PII is blocked, redacted, or rerouted.
4. A private strategy head can be evaluated without revealing source to NODE.
5. A high-sensitivity counterfactual lattice is committed before private selection feedback reaches NODE.
6. A counterfactual agency result cannot claim zero online selection leakage without a matching `CounterfactualLatticeReceipt`.
7. Private operators use the authenticated authority surface as the second logical party by default, unless policy explicitly chooses another threshold path.
8. A protected private-operator result cannot be admitted without a `PrivateOperatorReceipt`.
9. A broker action cannot be executed without a valid capability exit.
10. A restore cannot silently mutate local files without state-root and authority receipts.
11. A receipt proves private work without revealing protected plaintext.

22 Relation to Existing Work

cTEEs are not a replacement for FHE, MPC, ORAM, garbled circuits, or zero knowledge. They are a composition layer that chooses among them while preserving an agent-facing workspace abstraction. They can replace hardware confidential computing for workloads that fit the hardwareless replacement boundary: protected dependencies never enter the node as plaintext and are handled through no-plaintext-custody routing, cryptographic carriers, guardian/threshold evaluation, or capability exits.

FHE provides the strongest single-node cryptographic privacy story for functions that can be evaluated efficiently over ciphertext. MPC often provides more practical performance with a committee trust model. Garbled circuits fit boolean private logic and policy checks. ORAM mitigates access-pattern leakage. Hardware TEEs may allow plaintext inside an attested boundary, but introduce hardware trust and side-channel assumptions. cTEEs can use all of these, but their base claim is no-plaintext custody plus workload-level private operator routing, not universal private computation.

23 Open Problems

1. **Guardian minimization.** Treat the guardian as an authority view, not a new machine: authenticated browser, local agent harness, CLI signer, mobile passkey, enterprise key service, authority network, or threshold policy. Research should shrink this surface to decryption, private-head selection, budget accounting, and capability signing.

2. **Selection leakage.** Turn CLPD leakage into a measurable budget. The immediate path is fixed candidate counts, top- m or decoy reveal, delayed commit-reveal, schedule jitter, size padding, and cumulative leakage receipts.
3. **Counterfactual lattice efficiency.** CLE can suppress online selection leakage, but it spends additional public tokens. The key research problem is the coverage frontier: choose lattice width, depth, deduplication, and stopping rules that maximize probability of containing a private-optimal path per public token spent.
4. **Candidate-selection characterization.** Identify which agent tasks are candidate-selection-reducible, which require private generation, and which can be converted by redaction, public abstractions, or private heads. This is where the Private Agency Transform becomes an engineering science rather than a heuristic.
5. **Private transformer performance.** Reserve FHE/MPC transformer work for narrow private operators and small heads first. Use CLPD for default high-throughput agency until encrypted attention, sampling, and KV caches become practical enough for broader use.
6. **Encrypted developer tools.** Build diff, search, test, lint, and refactor tools that run in the authority view over locally decrypted objects, then commit encrypted patches and redacted summaries to the node.
7. **Private workspace UX.** Make unsafe plaintext mounts impossible by default and visually unmistakable when explicitly chosen. The normal product state should simply say: **Private Workspace On.**
8. **Receipt privacy.** Prove useful work occurred without revealing too much about the work. Near-term receipts should bind candidate lattice commitments, leakage budgets, redaction policy, and declassification outcomes.
9. **Private retrieval.** Combine encrypted names, neutral labels, coarse buckets, ORAM for high-sensitivity indexes, and local/guardian reranking so access patterns do not reveal the private workspace structure.
10. **Correctness for public candidates.** The node may generate poor or adversarial candidates. Use redundant public generation, deterministic replayable candidate commitments, verifier sampling, and dispute-triggered trace disclosure for settlement-grade work.

24 Conclusion

cTEEs make a simple claim technically precise:

Open Private Workspace. Run on untrusted compute. Do not give the node plaintext custody.

The contribution is not that software can magically hide plaintext from root. The contribution is a disciplined architecture for avoiding plaintext custody while preserving useful remote compute: encrypted workspace objects, public trunks, custody types, proof-carrying workspaces, the Private Agency Transform, Candidate-Lattice Private Decoding by default for protected agency, Counterfactual Lattice Execution when leakage budget requires it, sealed private heads, private workspace capsules, declassification gates, authority-bounded capability exits, receipts, and explicit leakage

profiles. This is enough to make persistent private agents practical on rented GPU infrastructure without pretending that the infrastructure is trusted. When a workflow instead sends sensitive plaintext to a third-party model API, the privacy claim changes: that path is provider trust unless the API receives only public, redacted, declassified, or separately protected inputs.

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